EXPLAINABLE AND BIAS-AWARE MULTI-LINGUAL SENTIMENT ANALYSIS WEB PLATFORM FOR INDIAN CODE-SWITCHED SOCIAL MEDIA TEXT

A mini-project synopsis report submitted by

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1. Project Title:

Explainable and Bias-Aware Multi-Lingual Sentiment Analysis Web Platform for Indian Code-Switched Social Media Text

2. Project Overview

India has over 600 million Internet users who mix Hindi , Tamil , Malayalam or Telegu with English which result in code-switched languages like Hinglish , Tanglish , Manglish and Tenglish , These code- switched languages holds a vital part in social media sentiment analysis still the major existing tools like Google NLP , AWS Comprehend or VADER are not built for code-switched languages , they tend to fail here , since they are completely built for monolingual text . We propose a full-stack web application that accurately classifies sentiment and emotion in all four Indian code-switched language pairs, explains predictions using SHAP token-level explainability, and audits model fairness across language groups — making it the first accessible, explainable, and bias-aware sentiment platform for Indian code-switched text.

3. Need of the Project

• **Accuracy Gap:** 700M+ Indians write in code-switched language. No existing tool reads it correctly — emotionally loaded words like bakwaas, romba, adipoli, baagundi are ignored or misclassified entirely.

• **Transparency Gap:** Current tools give predictions with no explanation — users cannot verify, trust, or challenge the result.

• **Fairness Gap:** No tool checks whether predictions are biased against Hindi-dominant vs Tamil-dominant vs English-dominant text. This bias silently exists in every system serving Indian audiences.

• **Platform Gap:** Research has produced models but no usable web platform. A marketing manager or health researcher cannot access these models without writing Python code.

4. Objective

Why: To address the critical gap where Indian code-switched social media text remains invisible to every existing AI sentiment tool, causing unreliable data-driven decision making for businesses, researchers, and healthcare workers.

What: To build the first unified, explainable, and fairness-aware sentiment analysis web platform supporting Hinglish, Tanglish, Manglish, and Tenglish.

How: To fine-tune XLM-RoBERTa on Indian code-switched datasets, integrate SHAP explainability, build a bias audit panel, and deliver a Flask web application with REST API and interactive dashboard.

Specific Objectives:

- To build a preprocessing and language detection pipeline for all four Indian code-switched language pairs.
- To fine-tune XLM-RoBERTa for 3-class sentiment classification (positive, negative, neutral) and 5-class emotion detection.
- To integrate SHAP token-level explainability revealing word-level sentiment drivers behind every prediction.
- To implement a bias audit panel measuring differential F1scores across Hindi-dominant, Tamil-dominant, and English-dominant text segments.
- To deliver a full-stack Flask web application with REST API, batch CSV processing, query history, and an interactive analytics dashboard.

5.Scope of the Project

Boundaries (In Scope):

- Sentiment analysis for Hinglish, Tanglish, Manglish, and Tenglish social media text.
- Single-text and batch CSV analysis via web interface and REST API
- SHAP token-level explainability for single-text mode.

- Bias audit dashboard comparing performance across language-dominant text groups.

SQlite query history with retrieval and deletion. Local deployment via Google Colab and ngrok.

Limitations (Out of Scope):

- Real-time social media streaming or live Twitter/Instagram data ingestion.
- Languages beyond the four identified pairs (e.g Kannada-English, Bengali-English) in this version.
- Mobile application, cloud deployment , or production-scale infrastructure.
- 100% accuracy – expect 70-82% on Hinglish, lower on Manglish/Tenglish due to limited training data.

6.Expected Outcome

- A fully functional web application accurately analysis sentiment and emotion in all four Indian Code-switched language pairs.
- REST API with endpoints for single-text analysis, batch CSV processing, and paginated query history.
- SHAP token-level explainability UI showing colour-coded word-level sentiment attribution.
- Bias audit dashboard for flagging performance gaps across Hindi-dominant, Tamil-dominant, and English-dominant text.
- Interactive analytics dashboard with sentiment distribution, emotion breakdown, and language mix charts.
- Documented open-source GitHub repository with source code, pipeline and demo video.

7.Beneficiaries

Who	Example	Benefit
E-commerce Platforms	Flipkart, Amazon India	Accurate Hinglish/Tanglish review analysis for product decisions
Brands & Marketers	FMGG, Telecom, Startups	Real Indian audience sentiment for campaigns and reputation monitoring
Film Industry	Kollywood, Tollywood	Early audience reaction in Regional-code-switched languages
Healthcare Orgs	NGOs, Hospitals	Detect distress signals in code-switched mental health posts.
Political Researchers	Think Tanks, Journalists	Genuine public opinion from Indian social media in native language.
Tourism Industry	Kerala, TN, AP Tourism	Accurate visitor feedback from Manglish and Tanglish reviews.
NLP Researchers	Universities, Students	First open baseline platform for Indian code-switched sentiment
700M Indian Users	Every Indian social media user	Their natural language finally understood and fairly evaluated by AI

Student Declaration

We hereby declare that the proposed project work is original and will be carried out under the guidance of the faculty mentor.

Student Signature(s): ,

Date: 01- 07- 2026

Faculty Guide Signature: